

Coding & Solution

Reference generation flow and prompt contract

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Team ID	SWTID-2026-9380
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Team	Lavinraj (Team Leader), Subash, Rahul, Hariharansudhan
Maximum Marks	5 Marks

Server-side generation pseudocode

```
function generatePlan(rawInput): profile = validateProfile(rawInput) prompt = buildPrompt(profile, PLAN_SCHEMA, SAFETY_RULES)
candidate = gemini.generate(prompt, timeout = 12s) parsed = parseStructuredOutput(candidate) plan = validatePlanSchema(parsed)
assertScheduleMatches(profile, plan) assertNoMedicalDiagnosis(plan) return sanitizeForDisplay(plan)
```

Prompt contract

Part	Instruction
Role	Act as a fitness-planning assistant, not a clinician.
Inputs	Use only the delimited, validated profile supplied by the application.
Output	Return JSON that exactly matches the provided plan schema; no extra prose.
Personalization	Respect goal, level, days, duration, equipment and preferences.
Safety	Avoid diagnosis and treatment; use cautious intensity; flag situations requiring qualified advice.
Quality	Include warm-up, cool-down, rest, technique cues, alternatives and a progression rule.
Failure	If safe personalization is not possible, return a structured limitation rather than inventing details.

Error handling

- Validation error: highlight the exact field and preserve the remaining answers.
- Timeout/provider error: keep the profile and offer a retry.
- Malformed response: retry once with stricter formatting, then show a safe failure state.
- Safety conflict: stop generation and display the appropriate professional-help message.